Mancala Heuristic Evaluation Report

Student ID: 2205117

Analysis of heuristic weight configurations and minimax search depth

1. Heuristic Descriptions

Heuristic	Very brief description
H1	Storage difference: stones in the player's store minus stones in the opponent's store.
H2	H1 plus a weighted difference in stones remaining on the two sides of the board.
H3	H2 plus a weighted difference in available extra moves.
H4	H3 plus a weighted difference in captures.
H5	Custom heuristic combining storage, board stones, extra moves, captures, extra-move quality, and capture-threat information.

2. Effect of Weight Configuration

Four weight configurations were evaluated. Each summary contains 96 games for H1-H4, while the fourth configuration also reports H5 over 128 games. The main observation is that changing weights can substantially alter the relative performance of the heuristics; there is no single weight configuration that makes every heuristic stronger.

Weights (w1,w2,w3,w4)	H1	H2	H3	H4
10, 1, 2, 4	44.79%	43.75%	47.92%	48.96%
10, 0.5, 1, 4	53.12%	37.50%	39.58%	58.33%
6, 1, 2, 6	48.96%	43.75%	43.75%	51.04%
10, 1, 2, 5	42.97%	45.31%	50.78%	58.59%

Average win rate across the four configurations: H1 = 47.46%, H2 = 42.58%, H3 = 45.51%, and H4 = 54.23%. Thus H4 is the strongest overall among H1-H4 in these experiments.

The first configuration is relatively balanced, with H4 only slightly ahead of H3. When the board-stone and extra-move weights are reduced (10, 0.5, 1, 4), H4 rises to 58.33% while H2 and H3 fall sharply. This suggests that the capture component of H4 is particularly useful when the contributions of the intermediate positional features are reduced. With (6, 1, 2, 6), increasing the capture weight again favors H4 (51.04%). The final configuration (10, 1, 2, 5) gives H4 its best result among the four common configurations, 58.59%, while H3 also reaches its best result, 50.78%.

H5 records 34.38% in the fourth experiment, considerably below H1-H4. Its larger feature set therefore does not automatically produce a better evaluation function. In particular, combining several features can introduce noisy or conflicting signals unless their scales and weights are carefully calibrated.

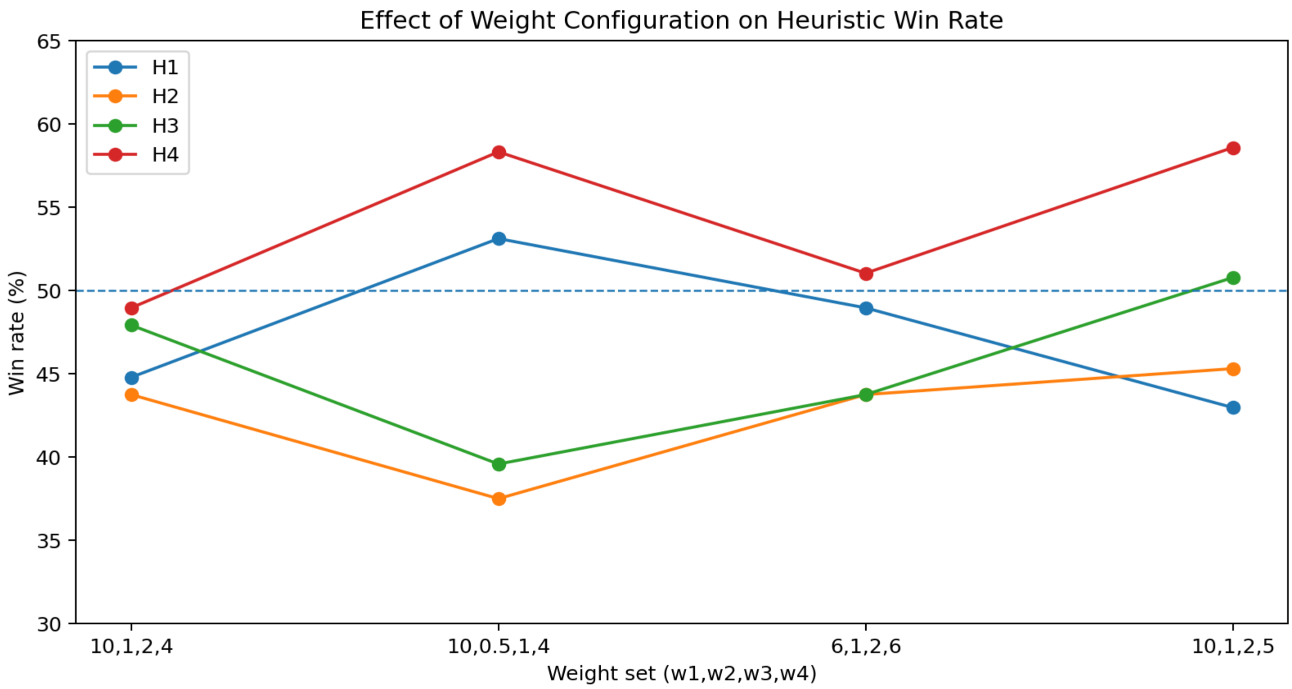

Figure 1. Win-rate changes across the four reported weight configurations.

3. Effect of Search Depth

The detailed experiment uses the reported weighting setup (10, 0, 5, 1, 4) and varies the minimax depths independently from 2, 4, 6, and 8. The results show that depth is not simply a property of a heuristic: the relative depth of the two competing players is crucial.

Clear depth advantage: For H1 versus H2, when H1 is fixed at depth 2 and H2 is increased through depths 2, 4, 6, and 8, H2 wins all four games with score margins of 20, 16, 2, and 24 points respectively (H1 minus H2 = -20, -16, -2, -24). However, when H1 is fixed at depth 8 and H2 is varied through 2, 4, 6, and 8, H1 wins all four games with margins of 26, 16, 14, and 28 points. This strongly indicates that deeper search can compensate for differences in heuristic quality.

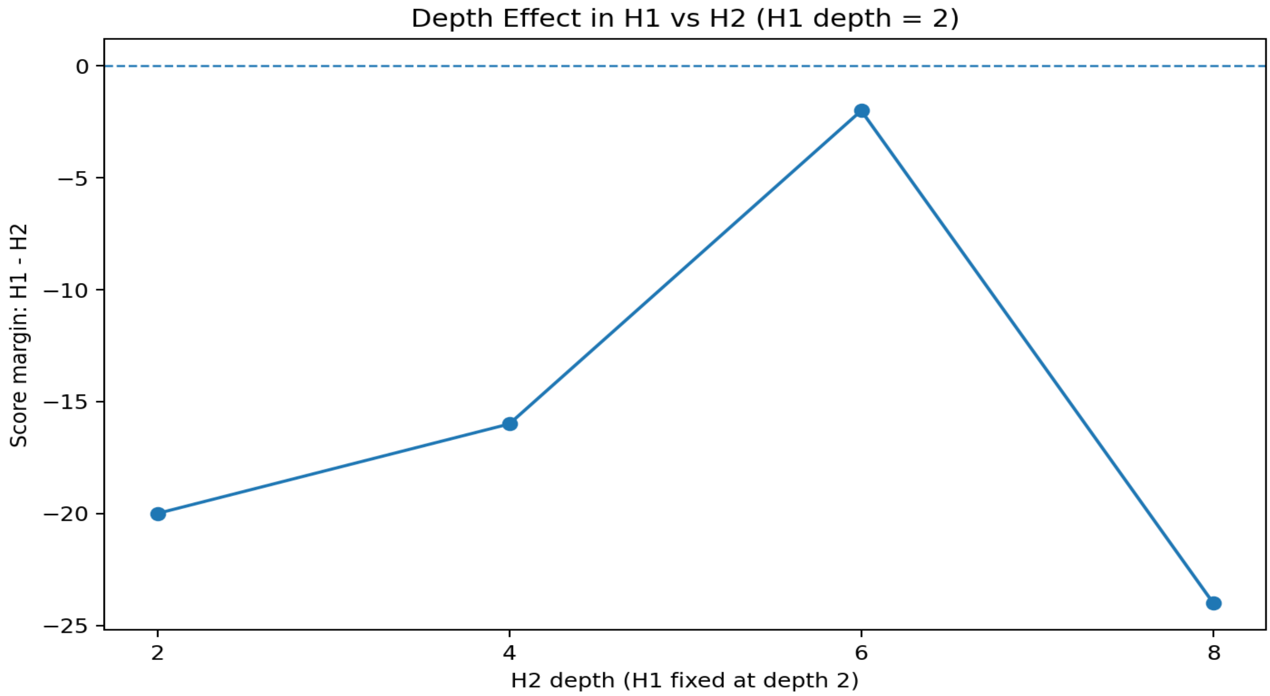

Figure 2. H1-H2 score margin when H1 remains at depth 2 and H2's depth changes.

The detailed results also show that the effect of depth is **non-monotonic**. For example, increasing an opponent's depth does not guarantee a progressively larger score difference at every depth. Mancala contains extra turns and captures, so a deeper search may discover tactical sequences that change the evaluation abruptly. Consequently, depth 8 is generally more capable than depth 2, but the benefit depends on the position, the opponent's depth, and the heuristic being used.

H4 is particularly robust in the detailed games. Against H2, for example, H4 at depth 2 defeats H2 at depths 2, 4, 6, and 8 with scores 30-18, 26-22, 35-13, and 29-19. Nevertheless, H4 is not invulnerable: when the depth advantage changes, some matchups become close or reverse, and several draws occur. This reinforces the conclusion that search depth and heuristic design interact rather than acting independently.

4. Overall Findings

• **H4 is the most consistently successful heuristic** in the weight summaries, averaging 54.23% wins and reaching 58.33% and 58.59% in two configurations.

• **Weight selection matters substantially.** H2 and H3 can lose performance when their feature weights are reduced, while H4 remains comparatively strong.

• **H5 does not outperform the simpler heuristics** in the supplied experiment. More evaluation features can add noise if their scales and interactions are not tuned.

• **Depth has a major effect.** The detailed results show that the deeper player often gains a substantial advantage, especially when comparing a depth-2 search with a depth-8 search.

• **Depth effects are not perfectly monotonic.** Tactical extra moves and captures can cause large changes between neighboring depth settings.

• **Heuristic quality and search depth should be evaluated together.** A weaker heuristic with substantially deeper search can outperform a stronger heuristic with shallow search.

5. Conclusion

The experiments indicate that the capture-aware H4 heuristic provides the best overall balance among the tested H1-H4 designs. Its performance is relatively stable across the different weight configurations and becomes especially strong when the capture feature receives a larger contribution. H5 demonstrates that simply adding more game features does not guarantee improved play; feature weighting and feature quality are both important.

The depth experiments further show that minimax depth can have an effect as large as, or larger than, the choice of heuristic. The strongest interpretation of the results is therefore not that one heuristic is universally best, but that good heuristic features, appropriate weights, and sufficient search depth must work together. For future experiments, a larger number of games per configuration and multiple random starting conditions would provide more reliable estimates of win rate and make the comparison statistically stronger.